

Lynbrook Math Summer Camp 2026

N1 Prime Factorization

Lynbrook Math Club

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N1.1 Lecture

Divisibility, Primes, and Composites

Before we dive into prime numbers, we must establish the language of divisibility. For integers x and y (where $x \neq 0$), we say x **divides** y if the quotient y/x is an integer. In this case, x is a **factor** of y , and y is a **multiple** of x . We typically denote this relationship as $x \mid y$.

Definition N1.1.1. An integer $p \geq 2$ is considered **prime** if its only positive divisors are 1 and itself. If an integer $n \geq 2$ has at least one positive divisor other than 1 and n , it is called **composite**.

Remark N1.1.2. The numbers 0 and 1 are classified as neither prime nor composite. The sequence of primes begins: 2, 3, 5, 7, 11, 13, 17, 19, ... Notice that 2 is the unique even prime! This tiny fact acts as a opening for many solutions.

Example 3 (Original)

Consider the sum of two prime numbers. If their sum is 101, what must the two prime numbers be?

Theorem N1.1.4 (Infinitude of Primes)

There are infinitely many prime numbers.

Euclid proved this elegantly: if we assume there is a finite list of primes $p_1, p_2, \dots, p_k$, the number $N = (p_1 p_2 \cdots p_k) + 1$ leaves a remainder of 1 when divided by any prime on our list. Thus, N must be divisible by some new prime not on our list, proving our assumption wrong.

Finding Primes and Algebraic Factoring

To test if a large number N is prime, we do not need to check divisibility by every number up to N .

Theorem N1.1.5

If an integer $N \geq 2$ is composite, it must have a prime factor p such that $p \leq \sqrt{N}$.

If N were the product of two numbers strictly greater than $\sqrt{N}$, their product would be greater than N , which is impossible.

Example 6 (Original)

To determine if 401 is prime, what is the largest prime number we must test as a potential divisor?

The Fundamental Theorem of Arithmetic**Theorem N1.1.7** (Fundamental Theorem of Arithmetic)

Every integer strictly greater than 1 can be expressed as a product of prime numbers in exactly one way, disregarding the order of the factors.

This is why primes are often called the "atoms" of mathematics. We typically write this **prime factorization** using exponents. For example, rather than writing $360 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 5$, we write $360 = 2^3 \cdot 3^2 \cdot 5^1$.

If a number is a perfect square, all exponents in its prime factorization must be multiples of 2. If it is a perfect cube, all exponents must be multiples of 3, and so on.

Example 8 (Original)

What is the smallest positive integer k such that $84k$ is a perfect square?

Legendre's Formula

Often, we need to find the prime factorization of very large numbers, specifically factorials like $n! = n \times (n-1) \times \cdots \times 1$.

Theorem N1.1.9 (Legendre's Formula)

For any positive integer n and prime p , the exponent of the largest power of p that divides $n!$ is given by:

$$E_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots$$

where $\lfloor x \rfloor$ is the floor function (round down to the nearest integer).

This formula simply counts how many multiples of p exist up to n (each contributing one factor of p), then adds the multiples of p^2 (which contribute a second factor of p), and so on.

Example 10 (Original)

Using Legendre's Formula, what is the highest power of 3 that divides $20!$?